Experiment 2

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Subject Code: 23CSH-382

1. **AIM:** To implement Single Page Application (SPA) navigation in the EcoTrack application using React Router, secure application routes using context-based authentication, and extend nested dashboard routing through follow-up enhancements.
2. **Objective:**
    a. Configure client-side routing in a React application using React Router
    b. Implement SPA navigation without full page reloads
    c. Design and apply protected routes using route-guard patterns
    d. Manage shared authentication state using React Context API
    e. Implement nested routing to build dashboard-style layouts
    f. Extend existing nested routes by adding new dashboard sections
    g. Implement logout functionality by updating shared context state
    h. Analyze route access behavior and explain redirection logic.
    i. Understand the role of Context API in shared state management and its comparison with Redux at an introductory level
3. **Implementation/Code:**
    a. **Routes.jsx**

```
import React from 'react'
import { useAuth } from '../context/auth'
import { Navigate } from 'react-router-dom';
```

```
export default function AuthProtectedRoutes({children}) {
   const {isAuthenticated} = useAuth();

   return isAuthenticated ? children : <Navigate to='/login' />
}
```

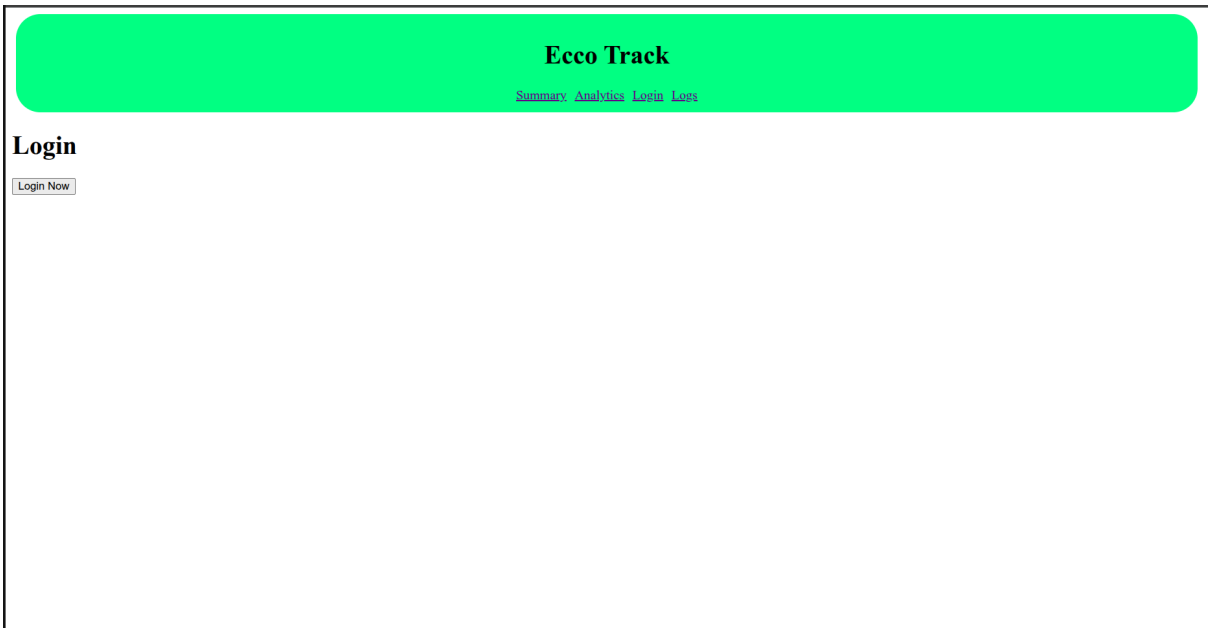
b. Dashboard_layout.jsx\

```
import React from 'react'
import { Outlet } from 'react-router-dom'

export default function Dashboard_layout() {
 return (
   <>
     <h1>Dashboard_layout</h1>

     <Outlet/>
   </>
 )
}
```

c. Coxtest.jsx

```
import { createContext, useContext, useState } from "react";

const AuthContext = createContext(null);

export const AuthProvider = ({children}) => {
   const [isAuthenticated, setIsAuthenticated] =
useState(false);

   return(
       <AuthContext.Provider value={{isAuthenticated,
setIsAuthenticated}}>
           {children}
       </AuthContext.Provider>
   )
}

export const useAuth = () => useContext(AuthContext);
```

4. Output:



Ecco Track

Summary Analytics Login Logs

Dashboard_layout

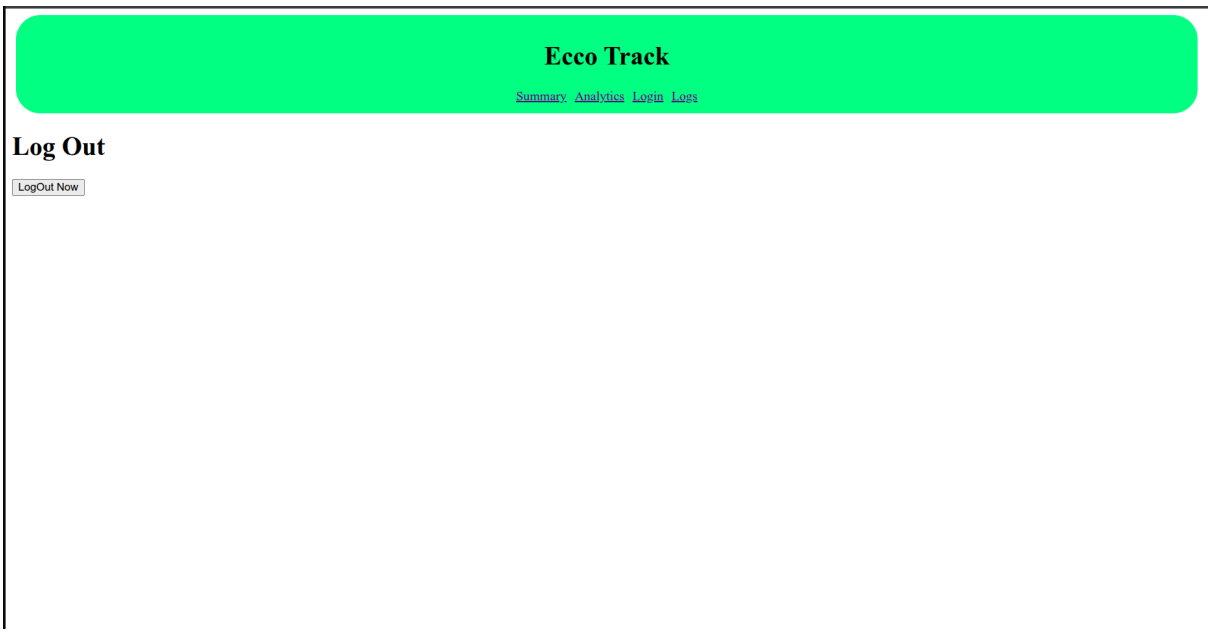
DashboardSummary

Ecco Track

Summary Analytics Login Logs

Dashboard_layout

DashboardAnalytics



5. Learning Outcome:

a. I learnt how to make react routes.
b. How the react routes actually works.
c. I learnt how to use the outlet tags along with nested routes.